Rocket Science and Technology 4363 Motor Ave., Culver City, CA 90232
Phone: (310) 839-8956 Fax: (310) 839-8855

# N.A.C.A Report 1307 Supplement for Downwash-Dependent Fin-Fin Interference (rev.2)

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by C. P. Hoult

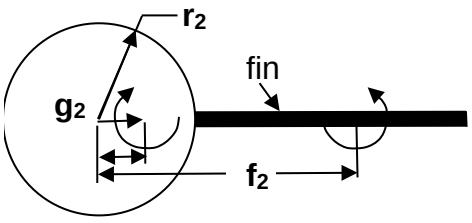
## Introduction

The justly famous NACA Report 1307[2] is the standard reference for estimating aerodynamic interference among the several components of a rocket. However, for a configuration with two, or more, sets of fins, 1307 is, in a practical sense, incomplete. The purpose of this memo is to document an approach to closing these gaps.

## Vortex Location

Fin-fin interference is estimated assuming that an upstream fin panel sheds a single horseshoe vortex with one trailing arm inside the rocket body and the other outside it. The outboard location of the exterior vortex arm is given by eq.(39) while the vortex strength **Γ** is given by eq.(38). The problem is then to locate the positions of the horseshoe vortex at the position of the after most fins, often mounted on a part of the body with a radius differing from that supporting the forward fins.

The sketch on the right shows a single second stage fin panel with its shed vortex located a distance $f_2$ from the centerline. The image vortex needed to satisfy the boundary condition on a cylinder of radius $r_2$ is located a distance $g_2$ from the centerline.




The key relationship[1] between these distances is

$$f_2 g_2 = r_2^2$$

Now, if the first stage body radius is $r_1 = r_2$, there is no issue. But, if this is not so, 1307 gives us only a gloomy excuse. Here is a quick, rough way to make the necessary adjustments to find $f_1$: Apply the incompressible continuity equation to the cross flow. Consider the air inside an annulus of outer radius $f_2$ and outside a radius $r_2$ at stage two. The mass flux in this annulus is

$$\frac{dm}{dt} = \rho A V = \pi\rho(f_2^2 - r_2^2)V ,$$

where $\frac{dm}{dt}$ = mass flux,

$\rho$ = mass density,

$A$ = stream tube area, and

$V$ = velocity.

Note that the shed vortex core lies on the outer surface of this annulus. Since the only velocity component normal to the axis of the annulus is an infinitesimal angle of attack times **V**, the vortex core will remain on the boundary of the same air mass as it moves aft. Assuming slender body aerodynamics, variations in $V$ will be small compared to $V$ itself during passage aft. It immediately follows that

$$f_1^2 - r_1^2 = f_2^2 - r_2^2, \text{ or}$$
$$f_1^2 = f_2^2 + r_1^2 - r_2^2.$$

Here the subscripts refer to the stage in question.

Fin Vortex Normal Force

Consider the common four-finned rocket configuration. The two fins normal to the fin in question contribute nothing to its normal force because their induced velocities cancel each other out. Induced velocities from the two vortices for the panel in question are supplemented by the induced velocities from the coplanar fin on the opposite side of the body. First, the external vortex at $y = f_1$ gives an induced velocity $v$ of

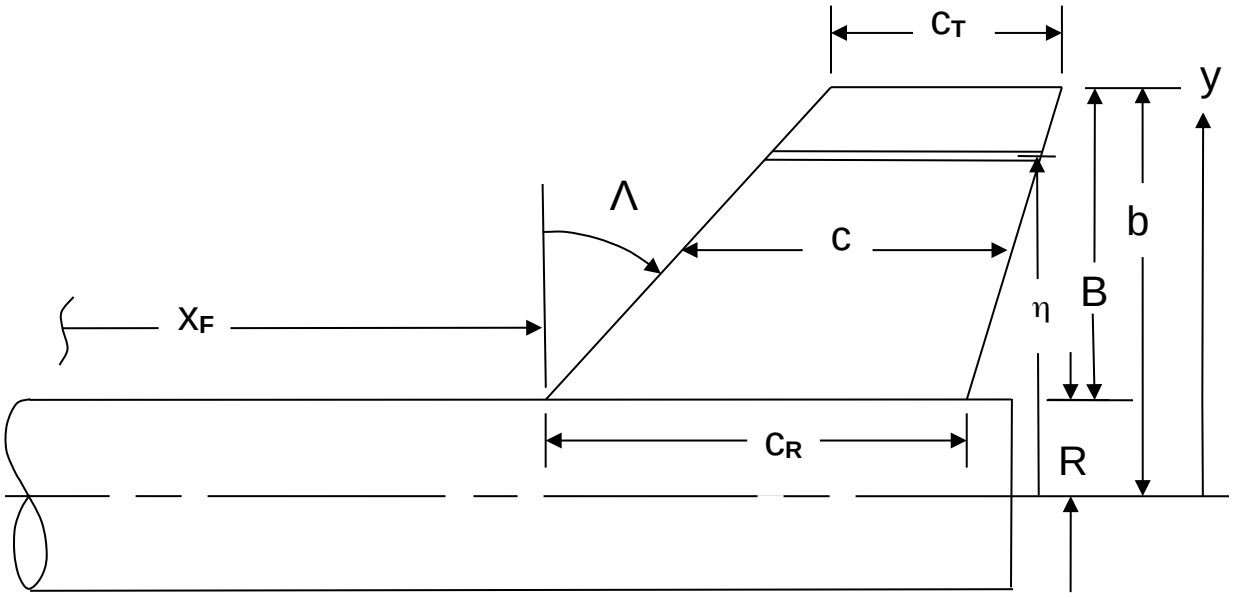
$$v = \frac{\frac{\partial \Gamma}{\partial \alpha}\alpha}{2\pi(y - f_1)}.$$

where $v$ = velocity induced normal to the fin plane, and
$\Gamma$ = vortex strength = circulation.

The local angle of attack is

$$\alpha = \frac{v}{V} = \frac{\frac{\partial \Gamma}{\partial \alpha}\alpha}{2\pi V(y - f_1)}.$$

We will use strip theory to estimate the normal force consequences of this local angle of attack. Consider the fin planform geometry:




The local chord $c$ is just

$$c(y) = \frac{c_R b - c_T R - (c_R - c_T)y}{(b - R)},$$

where the sketch serves to define the parameters.

Then, the vortex-induced normal force $dN$ acting on a first stage strip of span $dy$ is

$$dN = qc(y)dyc_{l_\alpha}\alpha = qcdyc_{l_\alpha}\frac{\frac{\partial\Gamma}{\partial\alpha}\alpha}{2\pi V(y - f_1)}, \text{ or}$$

$$dN = q\frac{c_R b - c_T R - (c_R - c_T)y}{b - R}dyc_{l_\alpha}\frac{\frac{\partial\Gamma}{\partial\alpha}\alpha}{2\pi V(y - f_1)}, \text{ or}$$

$$\frac{dN}{d\alpha} = \frac{qc_{l\alpha 1}\frac{\partial\Gamma}{\partial\alpha}}{2\pi V(b_1 - R_1)}\frac{c_{R1}b_1 - c_{T1}R_1 - (c_{R1} - c_{T1})y}{y - f_1}dy.$$

Here, $q$ = Dynamic pressure,

$c_{l_\varepsilon}$ = Two dimensional lift coefficient slope, and

$\alpha$ =.Angle of attack.

The subscripts "1" refer to the first stage fin parameters. At this point the reader is advised to proceed with caution: there's a singularity within the range of integration. Let's expand this to show the way out by using the principle value concept:

$$\int_{R_1}^{b_1}\frac{dy}{y - f_1} = \int_{f_1}^{b_1}\frac{dy}{y - f_1} + \int_{R_1}^{f_1}\frac{dy}{y - f_1} = \int_{f_1}^{b_1}\frac{dy}{y - f_1} + \int_{f_1}^{R_1}\frac{dy}{f_1 - y}$$

$$= \log(b_1 - f_1) - \log(f_1 - f_1) - \log(f_1 - R_1) + \log(f_1 - f_1) = \log(b_1 - f_1) - \log(f_1 - R_1).$$

This result follows because the two divergent logarithms are of the same order. With this, we can integrate from $R_1$ to $b_1$ to get

$$\frac{dN}{d\alpha} = \frac{qc_{l\alpha 1}\frac{\partial\Gamma}{\partial\alpha}}{2\pi V(b_1 - R_1)}\{[c_{R1}b_1 - c_{T1}R_1]\log[\frac{b_1 - f_1}{f_1 - R_1}] - [c_{R1} - c_{T1}][b_1 - R_1 + f_1\log[\frac{b_1 - f_1}{f_1 - R_1}]]\}$$

This, however, is only the part of the story reflecting the additional normal force acting on the panel next to the vortex. It remains to estimate the additional, negative, normal force $\Delta N$ acting on the panel on the opposite side of the rocket. For this panel, the angle of attack distribution is

$$\alpha = \frac{-\Gamma}{2\pi V(f_1 + y)}, \text{ and}$$

$$\frac{d\Delta N}{d\alpha} = -q\frac{c_R b - c_T R - (c_R - c_T)y}{b - R}dyc_{l\alpha 1}\frac{\frac{\partial\Gamma}{\partial\alpha}}{2\pi V(f_1 + y)}$$

Then, after integration,

$$\frac{d\Delta N}{d\alpha} = -\frac{qc_{l\alpha 1}\frac{\partial\Gamma}{\partial\alpha}}{2\pi V(b_1 - R_1)}\{(c_{R1}b_1 - c_{T1}R_1 + f_1(c_{R1} - c_{T1}))\log(\frac{b_1 + f_1}{f_1 + R_1}) - (c_{R1} - c_{T1})(b_1 - R_1\}\}$$

The sum, $N + \Delta N$, is the additional normal force induced on the stage 1 fins by one of the external trailing vortices. The image vortex result is similar except there is no singularity inside the integration region.

$$\frac{dN_I}{d\alpha} = -\frac{qc_{l\alpha 1}\frac{\partial\Gamma}{\partial\alpha}}{2\pi V(b_1 - R_1)}\{[c_{R1}b_1 - c_{T1}R_1 - (c_{R1} - c_{T1})g_1]\log[\frac{b_1 - g_1}{R_1 - g_1}] - [c_{R1} - c_{T1}][b_1 - R_1]\}$$

,

and

$$\frac{d\Delta N_I}{d\alpha} = \frac{qc_{l\alpha 1}\dfrac{\partial \Gamma}{\partial \alpha}}{2\pi V(b_1 - R_1)}\{(c_{R1}b_1 - c_{T1}R_1 + g_1(c_{R1} - c_{T1}))\log(\frac{b_1 + g_1}{g_1 + R_1}) - (c_{R1} - c_{T1})(b_1 - R_1)\}$$

.

The next step is to use these results to estimate a normal force coefficient by dividing the constant in front of the brackets by $qS_{\mathrm{Re}f}$:

$$C_{N\alpha} = \frac{c_{l\alpha 1}\dfrac{\partial \Gamma}{\partial \alpha}}{2\pi V S_{\mathrm{Re}f}(b_1 - R_1)}\{\}$$

According to ref.(2), eq.(38),

$$\frac{\dfrac{\partial \Gamma}{\partial \alpha}}{V} = \frac{K_{W(B)2}C_{L\alpha 2}S_{W2}}{4(f_2 - R_2)}.$$ Then,

$$C_{N\alpha} = \frac{S_{W2}}{S_{\mathrm{Re}f}}\frac{K_{W(B)2}C_{L\alpha 2}c_{l\alpha 1}}{8\pi(f_2 - R_2)(b_1 - R_1)}\{\}.$$

The average value of the 2D lift curve slope is, in fact, just the 3D value. Thus,

$$C_{N\alpha} = \frac{S_{W2}}{S_{\mathrm{Re}f}}\frac{K_{W(B)2}C_{L\alpha 2}C_{L\alpha 1}}{8\pi(f_2 - R_2)(b_1 - R_1)}\{\}$$

The total contribution from an image vortex is $N_I + \Delta N_I$, just as in the case of an external vortex. Add all four vortex terms, and multiply by 2 (to account for both fin panels) to the total first stage interference normal force. The remaining issue is the center of pressure of the vortex normal; force induced on the first stage fins. Assume that the vortex-induced normal force center of pressure is the same as for the basic normal force with body upwash.

The final vortex-induced normal force is that induced on the body behind the second stage fins. Report 1307, ref. (2), provides an estimate, eq. (38), for the vortex-induced body normal force. But, again, it's silent of the subject of the corresponding center of pressure. Assume that this center of pressure is the same as for the body normal; force due to the presence of a wing.

Fin Vortex-Induced Bending Moment

The downwash on the first stage fins also leads to additional bending moment on the fin. Proceeding as before, consider a single vortex at $y = f$. The induced bending moment from the other three can be found from the same formula. First,

$$M(y) = \frac{qC_{N\alpha 1}\Gamma}{2\pi V}\int_y^b c(\eta)\frac{\eta - y}{\eta - f}d\eta,$$

where

$$c(\eta) = \frac{c_R b - c_T R - (c_R - c_T)\eta}{(b - R)}.$$

The only trick comes when $b > f > y$; that is, when the vortex lies in the fin plane, and the evaluation station, $y$, can approach $f$. Then, near the vortex the integrand becomes indeterminate, essentially $0/\infty$.

But, to some extent, this is an artifact of the simplified analysis used. In the real world, the vorticity shed from the fore fins does not completely wrap up into pairs of concentrated vortices. Also, the fore fin wake does not remain perfectly coplanar with the aft fins. Instead, it usually is found above the aft fin plane. One could replace the troublesome denominator above by $\sqrt{(\eta - f)^2 + z^2}$, where $Z$ =the vertical displacement of the concentrated vortex. This would remove the troublesome pole at f, but still would not realistically model the distributed downwash field, especially near f. The plan is to go forward with the concentrated vortex keeping in mind the resulting bending moment is not valid near f, but would probably provide a satisfactory approximation near the fin root.

When the bending moment evaluation lies outboard of the vortex at $y = f$ there is no problem. We must evaluate integrals of the form:

$$\int_y^b \frac{d\eta}{\eta - f} = \log(b - f) - \log(y - f) = \log\frac{b - f}{y - f}$$

Both logarithms have positive arguments, and hence are well-behaved. But, when the bending moment evaluation station lies inboard of the vortex, we are formally asked to evaluate a logarithm having a negative argument, a non-starter. Then, the necessary trick is to break the integral into two parts:

$$\int_y^b \frac{d\eta}{\eta - f} = \int_{f+}^b \frac{d\eta}{\eta - f} - \int_y^{f-} \frac{d\eta}{f - \eta}$$

Here the $f+$ and $f-$ limits differ from $f$ by a small amount $\Delta f$. Then

$$\int_y^b \frac{d\eta}{\eta - f} = \log(b - f) - \log(f + - f) + \log(f - f -) - \log(f - y)$$
$$= \log(b - f) - \log(\Delta f) + \log(\Delta f) - \log(f - y)$$

In the limit as $\Delta f \to 0$ we find that

$$\int_y^b \frac{d\eta}{\eta - f} = \log(b - f) - \log(f - y) = \log\left|\frac{b - f}{y - f}\right|$$

which is valid for all $y$.

Then, if $A = \frac{c_R b - c_T R}{b - R}$ and $C = \frac{c_T - c_R}{b - R}$, the single vortex bending moment is

$$\frac{2\pi V}{q C_{N\alpha 1} \Gamma} M(y) = (-Ay + (A + Cy) f - Cf^2) \log\left|\frac{b - f}{y - f}\right| + (A + Cy)(b - y) - \frac{C}{2}\left[b^2 - y^2 + 2f(b - y)\right]$$

This result should be summed over all four vortices to obtain the overall bending moment. Finally, the bending moment functions $V(y)$ in ref. (3) are

$$V(y) = \frac{M(y)}{q C_{N\alpha 1}} = \frac{\partial\Gamma/\partial\alpha}{2\pi V}\{(-Ay + (A + Cy) f - Cf^2) \log\left|\frac{b - f}{y - f}\right| + (A + Cy)(b - y) - \frac{C}{2}[b^2 - y^2 + 2f(b - y)]\}$$

As before, this is only for a single vortex (not all four).

References


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3. Hoult, C. P., "Sounding Rocket Structural Design Loads (rev. 9)", 07 December 2013.